

24/09/26

Practical - 7. Sliding window

Aim:-

Write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL.

Sender Program:-

- * Input Window size from the user.
- * Input a Text message from the user.
- * Consider 1 character per frame.
- * Create a frame with following fields
[frame no., Data].
- * Send the frames.
- * Wait for the acknowledgement from the receiver.
- * Reader a file called Receiver-Buffer.
- * Check ACK field for the Acknowledgement no.
- * If ACK no. is as expected, send new set of frames accordingly. Else if NACK is received, resend the frames accordingly.

Create a receiver.

- * Reader a file called Sender-Buffer.
- * Check the Frame no.
- * If the frame no. are expected, write the appropriate ACK no. in the Receiver-Buffer file. Else write NACK no. in the Receiver-Buffer file.

Sender:-

import time

message = input ("Enter message: ")

window = int (input ("Enter window size: "))

i = 0

while i < len(message):

file = open ("Sender-Buffer ", "w")

print ("In Sending Frames: ")

for j in range (i, min (i+window, len (message))):

print ("Frame no: ", j+1, " DATA:", message[j])

file.write (str (j+1) + " " + message[j] + "\n")

file.close()

print ("Waiting for Ack. - ")

time.sleep(2)

file = open ("Receiver Buffer ", "r")

reply = file.readline().split()

file.close()

if reply[0] == "Ack":

print ("Ack received: ", reply[1])

i = int (reply[1]) - 1

elif reply[0] == "NACK":

print ("NACK received: ", reply[1])

i = int (reply[1]) - 1

print ("All frames sent successfully!")

Receiver:

```
file = open ("sender - Buffer", "r")
frames = file.readlines()
file.close()

Expected = 1
Print ("In Received Frames: ")

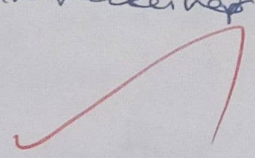
for line in frames:
    data = line.split()
    frame = int (data [0])
    Character = data [1]
    print ("Frame no: ", frame, "Data", Character)

    if frame == expected:
        print ("Frame accepted ")
        Expected = Expected + 1

        file = open ("Receiver - Buffer", "w")
        file.write ("ACK " + str (Expected))
        file.close()

    else:
        print ("Wrong frame!")
        file = open ("Receiver - Buffer", "w")
        file.write ("NACK " + str (Expected))
        file.close()

Print ("In Receiver finished.")
```



Q/P:

Q/P:

Enter message: HELLO

Enter window size: 3

Q/P:

Sending Frames:

Frame No: 1 DATA: H

Frame No: 2 DATA: E

Frame No: 3 DATA: L

Waiting for ACK...

Received frames:

Frame no: 1 Data: H

Frame accepted

Frame no: 2 Data: E

Frame accepted

Frame no: 3 Data: L

Frame accepted

ACK received: 4

Sending Frames:

Frame no: 4 Data: L

Frame no: 5 Data: O

Waiting for ACK...

Received Frames:

Frame no: 4 Data: L

Frame accepted

Frame no: 5 Data: O

Frame accepted

ACK Received: 6

All frames sent successfully!

Receiving Finished.

Result:-

Thus the sliding window program is executed successfully.